

Project Report: Smart Building Energy Optimization Assistant

Course: CISC 699-90 Applied Project

Name : Venkat Shyam Damagatla

Date: March 22, 2026

1. Project Overview

The **Smart Building Energy Optimization Assistant** is an intelligent system designed to balance the competing goals of energy efficiency and occupant comfort within a modern facility. By analyzing real-time data—including room schedules, weather forecasts, building occupancy, and device status, the system identifies opportunities to reduce energy consumption without compromising the environment for its users.

2. Introduction to Use Case Documentation

This report outlines the functional requirements of the system through ten distinct use cases. Each use case is documented following the strict guidelines of **Figure 4-7** and **Figure 4-8** from the course textbook.

- **Verb-Centric Actions:** Every use case name begins with a clear action verb.
 - **Standardized Format:** Each table includes Participating Actors, Entry/Exit Conditions, and Quality Requirements.
-

3. System Use Cases

UC4.1: Register Building Profile

Field	Details
Primary Actor	FacilityManager
Description	The manager sets up the initial building model, including zones and devices.
Participating Actors	Building Repository
Entry Condition	The FacilityManager is authenticated.

Field	Details
Flow of Events	1. The FacilityManager enters building zones, devices, and schedules. 2. The system validates the profile. 3. The system stores the building model. 4. The system confirms successful registration.
Exit Condition	A building profile is stored, or validation errors are returned.
Quality Requirements	Profile save completes within 4 seconds; Required fields are validated deterministically.

UC4.2: Import Occupancy Schedule

Field	Details
Primary Actor	FacilityManager
Description	The manager provides occupancy data to be linked with specific building zones.
Participating Actors	Occupancy Data Service
Entry Condition	A building profile exists.
Flow of Events	1. The manager uploads or enters occupancy data. 2. The system parses the schedule.

Field	Details
	3. The system links occupancy data to building zones. 4. The system stores the imported schedule.
Exit Condition	Occupancy schedule is stored, or format errors are reported.
Quality Requirements	Parsing errors identify exact rows; Import completes within 5 seconds.

UC4.3: Forecast Zone Demand

Field	Details
Primary Actor	Scheduler
Description	The system predicts energy demand based on environmental and occupancy data.
Participating Actors	Forecasting Service
Entry Condition	Required input data is available.
Flow of Events	1. The system gathers occupancy, weather, and device data. 2. The system forecasts energy demand by zone. 3. The system stores demand estimates.

Field	Details
	4. The system displays forecast values.
Exit Condition	Forecasts are stored, or missing inputs are reported.
Quality Requirements	Forecasts include timestamps and zone IDs; Processing completes within 6 seconds.

UC4.4: Recommend HVAC Setpoint Changes

Field	Details
Primary Actor	FacilityManager
Description	The system suggests energy-saving adjustments while considering comfort.
Participating Actors	Optimization Service
Entry Condition	Forecasts are available.
Flow of Events	1. The FacilityManager requests energy-saving recommendations. 2. The system evaluates forecasts and comfort constraints. 3. The system proposes setpoint changes by zone. 4. The system shows projected savings and comfort impact.
Exit Condition	Recommendations are displayed, or missing prerequisites are reported.

Field	Details
Quality Requirements	Savings returned as measurable values; Recommendations produced within 5 seconds.

UC4.5: Apply Approved Energy Plan

Field	Details
Primary Actor	FacilityManager
Description	The manager approves and executes a set of energy-saving commands.
Participating Actors	Device Control Gateway
Entry Condition	Approved recommendations exist.
Flow of Events	1. The manager approves selected recommendations. 2. The system packages the approved plan. 3. The system sends commands to the control gateway. 4. The system records command results.
Exit Condition	Commands are issued and logged, or failures are recorded.
Quality Requirements	Command status is returned per zone; Dispatch begins within 10 seconds.

UC4.6: Adapt Plan to Occupancy Change

Field	Details
Primary Actor	Occupancy Data Service

Field	Details
Description	The system recalculates energy plans when real-time occupancy shifts significantly.
Participating Actors	Optimization Service
Entry Condition	An active energy plan exists.
Flow of Events	1. The system detects an occupancy change. 2. The system compares the plan to the new context. 3. The system recalculates affected zones. 4. The manager is notified of revisions.
Exit Condition	Revised recommendations are stored or rationale for no change is provided.
Quality Requirements	Recalculation triggered within 2 seconds; Changed zones are identified.

UC4.7: Detect Comfort Violation Risk

Field	Details
Primary Actor	Scheduler
Description	The system identifies zones where energy plans might violate comfort thresholds.
Participating Actors	Comfort Monitoring Service

Field	Details
Entry Condition	An active or proposed plan exists.
Flow of Events	1. The system checks temperatures against thresholds. 2. The system identifies zones at risk. 3. The system generates alerts and mitigations. 4. The system records the alert.
Exit Condition	Comfort risk alerts are stored, or no risk is reported.
Quality Requirements	Risk score is measurable; Detection runs within 3 seconds.

UC4.8: Explain Recommendation

Field	Details
Primary Actor	FacilityManager
Description	The system provides a clear rationale for suggested energy actions.
Participating Actors	Explanation Service
Entry Condition	A recommendation exists.
Flow of Events	1. The manager selects a recommendation. 2. The system retrieves inputs and constraints.

Field	Details
	3. The system generates a concise explanation. 4. The system displays the explanation.
Exit Condition	Explanation is shown, or missing context is reported.
Quality Requirements	References energy and comfort factors; Generation completes within 4 seconds.

UC4.9: Generate Daily Savings Report

Field	Details
Primary Actor	FacilityManager
Description	The manager requests a summary of energy saved compared to the baseline.
Participating Actors	Reporting Service
Entry Condition	Energy usage data is available.
Flow of Events	1. The manager requests a daily report. 2. The system compares baseline and applied-plan usage. 3. The system generates savings metrics. 4. The system stores and displays the report.

Field	Details
Exit Condition	Daily report is saved, or incomplete data is reported.
Quality Requirements	Savings returned as numeric fields; Generation completes within 5 seconds.

UC4.10: Handle Sensor Data Outage

Field	Details
Primary Actor	Monitoring Service
Description	The system maintains operations using fallback estimates if sensors fail.
Participating Actors	Forecasting Service and FacilityManager
Entry Condition	Sensor data becomes unavailable.
Flow of Events	1. The system detects stale sensor data. 2. The system switches to fallback estimates. 3. Forecast confidence is marked as degraded. 4. The manager is notified.
Exit Condition	Fallback mode is activated or planning is paused.
Quality Requirements	Fallback status is explicit; Outage handling begins within 2 seconds.

4. System Requirements & Constraints

- **Performance:** Critical operations must trigger within **2–3 seconds**.

- **Security:** All management actions require authenticated access.
- **Reliability:** Fallback logic must trigger automatically during sensor outages.

5. Conclusion

By following the standards set in the course textbook, this report establishes a robust framework for the **Smart Building Energy Optimization Assistant**. These use cases ensure the final software product will be both energy-efficient and user-centric.